

Abstract No.

Decoding Aneuploidies in Glioma - using CRISPR technologies in mice to unravel novel cancer driver genes and vulnerabilities in recurrent CNA

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Aneuploidy, the accumulation of whole chromosome or chromosome arm imbalances, is one of the most prevalent alterations in cancer genomes. However, studying how aneuploidy contributes to tumorigenesis and identifying the driver genes on the altered chromosomes that confer a selective fitness advantage is notoriously difficult due to the vast number of genes affected. We previously developed a platform for multiplexed direct in vivo functional genomics knock-out screens (Loganathan et al., Science 2020; Yanchus et al., Science 2022; Langille et al., Cancer Discovery 2022). We refined our CRISPR strategy with a new CRISPR-activation methodology, termed CRISPR-KOALA (CRISPR-Knock Out and Activation Linked Assay), that enables high-throughput, multiplexed gene knockout and activation screens in sensitized mouse models of cancer. We focused on IDH-mutant Low Grade Glioma (LGGs) and GBMs and used KOALA to assess the transforming potential of all the genes residing on the most frequently altered chromosome arms. We uncovered several new and very potent driver genes on these chromosome arms, that converge on regulating distinct transcriptional pathways and provide insight into new potential therapeutic strategies for LGG and a novel template for multiplexed CRISPR screening directly in vivo.



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